

Arvinth Balaji

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SUMMARY

Electrical Engineering student at UC Santa Cruz with hands-on experience designing, building, and validating embedded AI systems, autonomous drones, and Formula SAE aerodynamic components from concept through tested prototype. Skilled across mechanical design, firmware, computer vision, PCB-level hardware, and technical documentation. Seeking an engineering internship to apply hardware and software development skills to real product challenges.

EDUCATION

University of California Santa Cruz, Santa Cruz, CA | September 2024 to Present

Bachelor of Science in Electrical Engineering, Expected Graduation May 2028

Relevant Coursework: Physics (Thermodynamics, Mechanics, Waves, Electricity and Magnetism), Chemistry, Computer Systems, Algorithms and Data Structures, Circuit Theory, Circuit Analysis and Design, Power Transfer, Analog Filters

SKILLS AND CERTIFICATIONS

Programming and Software: Python, C++, C, Java, JavaScript, R, SQL, MATLAB, Simulink, Bash, Jupyter Notebook, NumPy, Pandas, Matplotlib, Git, GitHub, Linux

Hardware Description and FPGA: Verilog, VHDL, FPGA Design, Digital Logic Design

Embedded and Hardware: Arduino, Raspberry Pi, STM32, ESP32, Microcontrollers, Embedded Systems, Firmware Development, Sensor Integration, Real-Time Systems, Edge Computing, Rapid Prototyping, Fabrication, Soldering, Breadboarding, I2C, SPI, UART, CAN Bus

Lab, Test, and Circuit Tools: Oscilloscope, Multimeter, Function Generator, Logic Analyzer, Power Supply, KiCad, Altium, Eagle, LTspice, SPICE Simulation

CAD and Mechanical Design: Onshape, SolidWorks, AutoCAD, Fusion 360, 3D Modeling, Design for Manufacturing (DFM), Design for Assembly (DFA), Tolerance Analysis, GD&T

Robotics and AI: Robot Operating System (ROS), Autonomous Systems, UAV Development, TensorFlow Lite, OpenCV, Computer Vision, Machine Learning, Deep Learning, Edge AI, Model Optimization

Data, Documentation, and Methods: Microsoft Excel, Microsoft Word, Microsoft Office, Google Sheets, Data Analysis, Statistical Process Control (SPC), LaTeX, Technical Documentation, SOP Writing, Circuit Analysis, Analog Circuits, Signal Processing, Power Electronics, Design Reviews (PDR/CDR), Root Cause Analysis, FMEA, Design of Experiments (DOE), Six Sigma, Lean Manufacturing

Certifications: Onshape Certified Associate (CAD), Certificate of Completion in Modular Prosthetics Research

RESEARCH

Modular Prosthetics Research Paper | Aug 2023 to Jan 2024

Researcher under the mentorship of Maria Fernanda-Lugo Bolaños

- Authored a publication-quality LaTeX research paper evaluating modular prosthetic systems across mechanical design, user adaptability, and long-term viability in clinical and low-resource environments, completing 5+ mentor-reviewed drafting cycles to meet scholarly rigor.
- Built quantitative cost and material feasibility models in Google Sheets and Microsoft Excel to produce evidence-based design recommendations, organizing performance metrics, statistical analysis, and data tables featured in the final published paper.

EXPERIENCE

Formula Slug Aero Team | April 2025 to Present

Team Member

- Owned end-to-end CAD design of aerodynamic components including front wing, diffuser, and sidepod through 3+ documented iteration cycles in Onshape and SolidWorks, advancing geometry from concept to competition-ready, manufacturable parts.
- Resolved cross-team packaging conflicts between aero, chassis, and electrical subteams in shared CAD assemblies, eliminating late-stage design rework before fabrication.
- Presented component geometry and material justifications at formal design reviews, replicating industry PDR and CDR processes and gaining hands-on Design for Manufacturing, tolerance analysis, and prototyping experience.

AI-Powered Camera System | Jan 2025

Personal Project

- Engineered a sub-50 ms real-time computer vision system integrating Arduino microcontrollers with Python and C++ for live object detection and motion tracking on resource-constrained embedded hardware, communicating between modules over I2C, SPI, and UART.
- Designed the full mechanical enclosure in Onshape and tuned lightweight machine learning models in TensorFlow Lite and OpenCV for edge deployment, benchmarking accuracy, latency, and power draw across firmware iterations using oscilloscope, multimeter, and logic analyzer measurements under reproducible test protocols.
- Documented a scalable system architecture portable to robotics, UAV, and security applications including Raspberry Pi, STM32, and ESP32 platforms, validating signal integrity through SPICE simulation and structured debugging.

Arduino Seedling Drone | Jun 2024 to Aug 2024

Researcher, Designer, and Builder

- Designed the airframe in Onshape, fabricated the structure, soldered and bench-tested the electronics, and integrated a multi-sensor camera payload covering color, shape, and thermal imaging, delivering a fully functional custom drone solo in 8 weeks from blank CAD canvas to field-tested prototype.
- Built a Python and OpenCV computer vision pipeline using NumPy and Pandas to classify planting zones from real-time aerial imagery by vegetation density and soil temperature thresholds, enabling autonomous field mapping without manual annotation.
- Managed the full product lifecycle independently across concept, CAD, firmware in Arduino IDE, sensor integration over I2C and SPI, field testing, and documentation, demonstrating end-to-end ownership expected of an engineering intern.